

PREETHI AMASA

Software Engineer, AI Focused

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PROFESSIONAL SUMMARY

Software engineer with full-stack production experience (React, Next.js, Node.js, TypeScript, ASP.NET, FastAPI) and hands-on work building autonomous, tool-using AI agent systems (OpenAI Agents SDK, LangGraph, CrewAI, AutoGen, MCP). Comfortable owning a feature end to end from UI and API design to backend logic and deployment and equally comfortable integrating LLM-driven components into production systems. Uses AI-native tooling (Claude Code, Cursor) as part of the daily engineering workflow.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#

Frontend: React, React Native, Next.js, Redux, Tailwind CSS, HTML5, CSS3

Backend & APIs: Node.js, FastAPI, ASP.NET, Microservices, Firebase, Supabase

Databases: PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS (Lambda, S3), Serverless Architecture, Docker, GitHub Actions, CI/CD, Git, Linux

AI & Agentic Tooling: OpenAI Agents SDK, LangGraph, LangChain, CrewAI, AutoGen, MCP, RAG, Claude Code, Cursor, Prompt Engineering

Tools & Testing: Postman, Agile, Slack, Jira, Confluence

EXPERIENCE

The Citizen Project

Software Engineer - AI

Jul 2026 - Present

New York, NY

- Ship features on a production Next.js/AWS Amplify application within an Agile/Scrum team, coordinating via Slack and Confluence.
- Lead the rollout of AI coding agents into CI/CD and QA workflows to streamline the engineering process.
- Manually test and document each feature and ticket fix prior to release, and support SRE work by triaging production issues and auditing environment configs.

Technologies: Next.js, React, Node.js, AWS Amplify, Git, Jira, Confluence, Slack

South Dakota State University

AI Research Engineer, Vision-Language Models

Aug 2025 - Jun 2026

Brookings, SD

- Designed RegiGrow, a continual learning framework combining Mixture-of-Experts LoRA adapters with an automated, entropy-based growth mechanism that lets the model expand its own capacity as needed rather than on a fixed schedule.
- Built the training infrastructure supporting this research, including automated pipelines that processed 45,000+ video frames across HPC GPU clusters to fine-tune a frozen ImageBind embedding encoder.
- Evaluated the framework on the ShanghaiTech, UCF-Crime, and XD-Violence video anomaly and violence-detection benchmarks to validate performance against established baselines.
- Applied Claude Code and Cursor for pipeline development, prompt iteration, and test generation, reducing development time on training scripts.

Technologies: Python, PyTorch, LoRA, Claude Code, Cursor, Linux, SLURM, Git

South Dakota State University IT Division

Software Engineer, Backend and Full Stack

Jan 2025 - Aug 2025

Brookings, SD

- Led end-to-end development of 15+ production REST API features and web/console applications, from requirements and design through deployment, using C#, ASP.NET, Blazor, and SQL Server, supporting 1,000+ active users.
- Built dynamic data collection and reporting workflows via the Formsite and SharePoint APIs, and automated backend pipelines with iPaaS for scheduled and event-driven processes.
- Optimized SQL queries, joins, and CTEs to cut execution time by 40%, and introduced code-review validation improvements that reduced production errors by 30%.
- Manually tested and documented each feature and ticket resolution, producing technical documentation for codebases, APIs, and workflows that reduced onboarding time, while providing ongoing troubleshooting support for deployed systems.

Technologies: TypeScript, React, C#, ASP.NET, Blazor, SQL Server, Formsite API, SharePoint API, iPaaS, Git, Agile

- Built automated SQL and Python pipelines for reporting processes previously performed manually, reducing manual effort by 40%.
- Performed feature engineering and statistical analysis using Pandas and NumPy across 20+ stakeholder reporting requests.

Technologies: Python, SQL, Pandas, NumPy, Git

PROJECTS

MCP Security Gateway: policy-enforced proxy against indirect prompt injection [Live Demo](#)

Architected a full-stack security gateway that intercepts every AI agent tool call (calendar, file, email) to defend against indirect prompt injection a vulnerability where a hidden instruction in tool-returned data (e.g., a poisoned calendar event) hijacks an agent into leaking data or sending unauthorized emails without direct attacker contact. Built a deliberately unprotected reference agent to demonstrate the exploit live, then engineered a FastAPI/Pydantic policy-enforcement layer that authorizes every tool call against a per-agent permission table before execution. Designed a two-layer detection pipeline heuristic pattern matching backed by an LLM classifier (GPT-4o-mini) for paraphrased attacks with full audit logging to Postgres (Supabase) capturing every decision and its reasoning. Shipped a live Next.js/TypeScript dashboard (Vercel + Render) with a real-time call feed, policy manager, and an on-demand attack simulator that fires live attacks through the gateway and shows them blocked in real time.

Stack: Python · FastAPI · Pydantic · Uvicorn · OpenAI API (function calling) · Supabase/Postgres · Next.js · TypeScript · Tailwind CSS · Vercel · Render

AI Agents: Multi-Agent Orchestration [GitHub](#)

Built 8 autonomous, tool-using agent systems spanning the OpenAI Agents SDK, CrewAI, LangGraph, AutoGen, and MCP, including a Career Digital Twin, an SDR agent that autonomously writes and sends outreach emails through SendGrid, a CrewAI Stock Picker, a four-agent Docker-based Engineering Team capable of writing, testing, and maintaining its own code, and an AutoGen Agent Creator that generates new agents from a single prompt. Delivered a Trading Floor capstone coordinating 4 agents, 6 MCP servers, and 44 tools across 5 LLM providers.

Stack: OpenAI Agents SDK · CrewAI · LangGraph · AutoGen · MCP · Docker · SendGrid

Regret.log: anonymous, categorized regrets platform [regrets.log](#)

Built a full-stack app where users anonymously post categorized regrets, browse a public feed filterable by category and sortable by recency or reactions, and react to posts with live emoji counts. Implemented Google OAuth sign-in (used only for spam prevention, nothing shown publicly) and Row Level Security across all data access in Supabase, and shipped a personal dashboard with avatar selection, post history, and reaction totals; deployed live on Vercel.

Stack: Next.js (App Router) · React · TypeScript · Tailwind CSS · Supabase (Postgres, OAuth, RLS) · Vercel

EDUCATION

South Dakota State University - MS, Computer and Information Sciences, GPA 3.7	May 2026
Annamacharya Institute - B.Tech, Artificial Intelligence and Data Science, GPA 3.9	May 2024

CERTIFICATIONS

- AWS Certified Data Engineer - Associate (2026)
- AI Agents with OpenAI Agents SDK, CrewAI, LangGraph, AutoGen & MCP
- IBM Introduction to DevOps
- IBM Continuous Integration and Continuous Delivery (CI/CD)